Paper 7.75 - Discrete-Continuous Bridge as Next-State Precondition

CQE-paper-07.75
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Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-07.75.md

Paper 7.75 - Discrete-Continuous Bridge as Next-State Precondition

Purpose

Paper 7.75 explains how sample-preserving bridges become preconditions for Paper 8 and later continuum papers.

Exported State

Paper 7 exports:

```
indexed trace
piecewise-linear bridge
zero sample error at integer points
endpoint agreement
original discrete receipt
scope boundary for between-sample dynamics
```

Use in Paper 8

Paper 8 may use bridge outputs as sampled receipts entering a lattice closure template. It must preserve the indexed samples and avoid treating the interpolant as a proof of lattice closure.

The allowed transition is:

```
sample-preserved trace -> lattice closure input
```

The disallowed transition is:

```
continuous drawing -> closed lattice theorem without receipt
```

Use in Paper 9

Paper 9 may use indexed traces as Hamiltonian window inputs. If it claims time or dynamics beyond sample preservation, Paper 9 must supply its own verifier.

Use in Paper 16

Paper 16 may read residuals at continuum edges only if it keeps the Paper 7 scope boundary visible. Continuum collapse is not proved by interpolation.

Precondition Rule

Before a later paper uses Paper 7, it should be able to answer:

```
What discrete trace is being bridged?
Are all indexed samples preserved?
What is the sample error?
Is any between-sample claim being made?
Which receipt proves the bridge?
```

Conclusion

Paper 7.75 turns sample-preserving interpolation into portable state. It lets later papers draw and transport traces while keeping the discrete proof and scope boundary intact.